

ALEXANDRIA, MN, USA

WMO: 726557

Lat: 45.868N

Lon: 95.394W

Elev: 1416

StdP: 13.96

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14910

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.8	-12.8	-23.7	1.6	-16.8	-18.9	2.1	-11.8	28.1	13.6	25.8	16.7	8.0	290	0.653

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	88.1	72.2	84.9	70.3	82.2	68.8	75.4	84.1	73.3	81.5	71.2	79.1	13.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.5	127.1	81.1	70.5	118.3	78.8	68.4	110.0	76.4	39.9	84.1	37.8	81.4	35.9	79.4	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.8	23.2	20.4		-23.4	93.1	5.9	3.0	-27.6	95.2	-31.1	97.0	-34.4	98.7	-38.7	100.8	
				DB												
				WB	-23.3	78.6	5.0	1.9	-26.9	79.9	-29.9	81.1	-32.7	82.1	-36.4	83.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.9	10.8	14.7	27.8	42.9	56.0	66.1	71.0	68.6	60.6	46.3	30.9	17.3
	DBStd	23.42	14.07	13.03	13.05	10.36	8.69	6.79	5.62	5.34	8.33	9.17	11.01	12.31
	HDD50	4976	1216	988	693	255	39	1	0	0	11	182	577	1014
	HDD65	8576	1681	1408	1153	663	302	65	14	25	181	583	1022	1479
	CDD50	2386	0	0	5	43	225	484	652	577	328	67	5	0
	CDD65	513	0	0	0	1	23	99	201	137	49	3	0	0
	CDH74	4132	0	0	0	20	271	805	1639	980	383	34	0	0
	CDH80	1105	0	0	0	4	80	214	480	234	88	5	0	0
(14) Wind	WSAvg	10.4	10.6	10.5	10.8	11.7	11.2	9.7	9.0	8.8	10.0	11.2	11.1	10.6
(15) Precipitation	PrecAvg	25.6	0.9	0.6	1.3	2.3	3.0	4.3	3.1	3.3	2.5	2.1	1.2	0.7
	PrecMax	40.0	4.2	2.2	3.0	6.5	5.9	10.9	8.0	8.0	5.4	8.2	4.1	2.6
	PrecMin	11.4	0.0	0.0	0.2	0.0	0.3	0.6	0.8	0.2	0.4	0.0	0.0	0.1
	PrecStd	6.1	0.8	0.4	0.7	1.4	1.3	2.3	1.7	2.1	1.5	2.0	1.1	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.1	47.5	64.8	76.1	88.4	91.5	92.4	90.0	87.2	79.2	63.7	45.9
		MCWB	36.4	43.3	54.1	58.9	65.6	72.5	76.0	73.3	70.5	62.4	53.3	40.5
	2%	DB	36.3	40.5	57.1	69.3	81.3	85.4	88.1	85.6	82.3	70.9	56.5	39.2
		MCWB	32.9	37.0	48.8	52.9	63.7	70.0	73.8	71.5	68.4	58.4	47.7	35.3
	5%	DB	33.0	36.2	50.2	64.6	75.8	81.7	85.1	82.4	78.6	64.7	50.5	35.6
		MCWB	31.3	33.7	44.3	50.6	60.3	67.5	71.5	69.4	66.1	55.3	44.7	32.9
	10%	DB	30.1	33.3	44.3	59.8	71.2	78.5	82.3	79.7	74.5	60.3	45.8	32.7
		MCWB	28.5	31.6	40.1	48.3	58.2	65.7	70.0	67.8	64.1	52.5	40.9	31.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.9	43.2	56.9	60.2	70.0	75.9	78.9	77.1	74.0	64.4	55.8	41.6
		MCDB	41.5	47.3	61.7	73.4	82.2	86.1	87.7	86.5	82.3	72.5	61.4	44.8
	2%	WB	33.4	37.2	50.1	55.5	65.8	72.7	76.5	74.0	70.9	60.5	49.3	35.7
		MCDB	35.4	40.0	55.5	64.5	76.6	82.1	85.3	81.8	78.9	68.2	55.1	38.5
	5%	WB	31.3	33.9	44.4	52.0	62.6	70.1	74.2	71.9	68.6	56.6	45.0	33.3
		MCDB	33.1	35.8	49.8	62.3	72.5	79.4	82.5	79.2	75.8	63.6	50.2	35.3
	10%	WB	28.5	31.2	40.0	49.2	59.5	67.7	71.9	69.9	66.1	53.2	41.2	31.1
		MCDB	30.0	33.2	44.4	58.5	68.8	75.9	79.6	77.0	72.6	58.8	46.0	32.8
(35) Mean Daily Temperature Range	MDBR		15.8	16.6	16.2	18.6	19.4	18.2	18.2	18.4	18.6	16.2	13.9	13.7
	5% DB	MCDBR	16.7	17.8	22.7	27.2	27.1	22.1	20.8	21.5	23.2	23.7	21.3	17.1
		MCWBR	14.2	14.9	15.1	14.9	13.6	10.8	10.1	10.4	11.8	13.6	14.1	13.9
	5% WB	MCDBR	15.7	16.7	22.0	24.6	23.2	19.4	19.2	18.9	19.5	21.1	20.0	15.5
		MCWBR	13.6	14.1	15.2	14.9	13.4	11.0	11.1	11.0	11.6	13.5	13.9	13.0
(40) Clear-Sky Solar Irradiance	taub		0.262	0.285	0.305	0.356	0.381	0.382	0.380	0.387	0.354	0.308	0.278	0.255
	taud		2.397	2.335	2.404	2.337	2.326	2.372	2.377	2.355	2.432	2.543	2.528	2.443
	Ebn at Noon		272	285	294	286	281	281	279	273	274	275	266	263
	Edh at Noon		22	29	32	37	39	37	37	36	31	24	20	19
(44) All-Sky Solar Radiation	RadAvg		534	883	1208	1510	1711	1891	2019	1688	1280	803	507	370
	RadStd		66	98	113	126	129	125	77	142	96	119	50	48

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page